

Kian (Mohammad) Kianpisheh

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🌐 <https://kianpisheh.github.io/>

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Summary: I'm a PhD candidate in Computer Science at the University of Toronto, supervised by Prof. Khai Truong in the DGP lab. My research focuses on ubiquitous computing and applied machine learning, designing intelligent systems that adapt to users' habits, preferences, and environments.

EDUCATION

University of Toronto

Sep 2018-Present

Ph.D. candidate in Computer Science, DGP lab

Personalized Ubiquitous Intelligent Systems

Advisor: Prof. Khai Truong

Sharif University of Technology

2015-2017

M.Sc. in Electrical Engineering, (Computer Vision, Scene Understanding)

Advisor: Prof. Mohammad Sharifkhani

Shahed University

2008-2013

B.Sc. in Electrical Engineering, (Image Processing)

Advisor: Prof. Alireza Behrad

EXPERIENCE

University of Toronto

2018-2025

Research Assistant — Applied AI

- Fine-tuned lightweight LLMs (Phi-3.5 Mini, and TinyLlama) using HuggingFace Transformers to power a context-aware human activity recognition agent.
- Leveraged user feedback as a Retrieval-Augmented Generation (RAG) resource to personalize human activity recognition and improve predictions with Hugging Face Transformers.
- Designed personalized intelligent systems using multimodal machine learning techniques, including self-supervised and one-shot learning, for adaptive behavior modeling.
- Worked with motion, vision, audio, and other sensor data to model human behavior and context.
- Focused on adaptive and user-aware AI/IoT applications, combining signal processing with deep learning approaches.

Mathsronauts

2023-2025

Machine Learning Engineer (PyTorch, OpenCV)

- Developed a platform that allowed students to interactively learn and experiment with computer vision and machine learning concepts.
- Built the user interfaces for the teaching platform for students using React.js, TypeScript, and Redux.
- Implemented backend APIs using Node.js and Express.js to support user authentication and dynamic content delivery.

Paya Vision

2013-2015

Machine Learning Engineer (TensorFlow, OpenCV, OpenCL, Scikit-Learn)

- Developed a content-based video retrieval system based on Topic Models for traffic surveillance videos.
- Developed an anomaly detection system for traffic surveillance videos using Topic Models.
- Designed and implemented a license plate recognition system.

RESEARCH AND PUBLICATIONS

Kianpisheh M, Mariakakis A, Truong KN. SAHARA: Self-supervised Approach for Human Activity Recognition based on Everyday Audio Events. (under review)

Kianpisheh M, Mariakakis A, Truong KN. exHAR: An Interface for Helping Non-Experts Develop and Debug Knowledge-based Human Activity Recognition Systems. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies. 2024 Mar 6;8(1):1-30. [\(link\)](#)

Kianpisheh M, Li FM, Truong KN. Face recognition assistant for people with visual impairments. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies. 2019 Sep 9;3(3):1-24. [\(link\)](#)

Kianpisheh M, Content-based Video Retrieval in Traffic Videos using Latent Dirichlet Allocation Topic Model. 2017 Sep, arxiv. [\(link\)](#)

SELECT LIST OF TECHNICAL PROJECTS

SAHARA: An audio-based self-supervised human activity recognition system. 2025
The system automatically clusters repetitive audio events and characterizes the user's activity based on those detected events.

IMUdio: A smartwatch human activity recognition system based on audio and IMU sensor data. 2024
A lightweight deep learning model that fuses audio and IMU data embeddings to recognize human activities, with inference performed directly on a smartwatch.

exHAR: An Interface for Helping Non-Experts Develop and Debug Human Activity Recognition Systems. 2024
An explainable framework that enables users to personalize sensor-based human activity recognition systems.

SensorLab: A data visualization tool for wearable devices and smartphone IMU sensor data. 2023
A visualization web application that shows the spectrogram of IMU (i.e., accelerometer and gyroscope) sensor data.

ActivityLab: A data visualization tool for wearable devices and smartphone sensors data. 2022
A web-based visualization application that depicts human activity datasets as a chronological sequence of events. Users can program the system in If-This-Then-That format.

Face recognition assistant for people with visual impairments. 2020
A smart wearable camera that automatically captures the faces of people the user interacts with. It filters out low-quality images (poor lighting, extreme head poses, motion blur) to ensure reliable face classification.

CBVR: Content-based Video Retrieval using Latent Dirichlet Allocation Topic Model.

2017

The system detects traffic patterns using Topic Models. A GUI allows users to search for specific patterns efficiently.

TEACHING

University of Toronto

Foundation of Computer Vision

Programming on the Web (Lead TA) (4x)

Introduction to Computer Science (>4x)

Computational Thinking

Image Understanding (2x)

Data Structures and Analysis

Introduction to Programming (>4x)

Great Ideas in Computing

SKILLS

Machine Learning: PyTorch  TensorFlow  Hugging Face  OpenCV 

Web development: React  React  Node  JavaScript  TypeScript 

Design: Figma  Illustrator 

Mobile Development: Android (Java) 

REFERENCES

Khai N. Truong (Professor, Department of Computer Science, University of Toronto)

Alex Mariakakis (Asistant Professor, Department of Computer Science, University of Toronto)